

BACKLOG BUDDY ? STUDY MATERIAL

Geotechnical Engineering ? Unit-wise Lecture Notes

Subject Code: CE401 Semester: 4

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Condensed notes covering all five units: Soil Properties and Classification, Permeability and Seepage, Compaction and Consolidation, Shear Strength of Soils, Bearing Capacity and Settlement.

Unit 1: Soil Properties and Classification

- Definition, key concepts and significance of Soil Properties and Classification
- Important diagrams / derivations / algorithms for Soil Properties and Classification
- Solved examples and previous-year questions on Soil Properties and Classification

Unit 2: Permeability and Seepage

- Definition, key concepts and significance of Permeability and Seepage
- Important diagrams / derivations / algorithms for Permeability and Seepage
- Solved examples and previous-year questions on Permeability and Seepage

Unit 3: Compaction and Consolidation

- Definition, key concepts and significance of Compaction and Consolidation
- Important diagrams / derivations / algorithms for Compaction and Consolidation
- Solved examples and previous-year questions on Compaction and Consolidation

Unit 4: Shear Strength of Soils

- Definition, key concepts and significance of Shear Strength of Soils
- Important diagrams / derivations / algorithms for Shear Strength of Soils
- Solved examples and previous-year questions on Shear Strength of Soils

Unit 5: Bearing Capacity and Settlement

- Definition, key concepts and significance of Bearing Capacity and Settlement
- Important diagrams / derivations / algorithms for Bearing Capacity and Settlement
- Solved examples and previous-year questions on Bearing Capacity and Settlement

Exam Tips:

- * Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- * Solve at least 3 previous year papers under timed conditions before the exam.
- * Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- * Revise definitions, diagrams and standard derivations ? they earn easy marks.
- * Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

Appearance counts are based on available papers and are not exam predictions.